

Homework 3

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Exercise 1

Notice that

$$I(T) = \int_0^T W^6(t) dW(t)$$

is an Ito integral. Hence,

$$E[I(T)] = 0$$

and

$$\begin{aligned} \text{Var}[I(T)] &= E[I^2(T)] - (E[I(T)])^2 \\ &= \int_0^T E[W^{12}(t)] dt \end{aligned}$$

by Ito isometry.

By definition, $W(t) \sim \mathcal{N}(0, t)$ and the characteristic function $\phi_{W(1)}(u) = e^{-\frac{1}{2}u^2}$. Mathematica yields that

$$E[W^{12}(1)] = \frac{d^{12}}{du^{12}} e^{-\frac{1}{2}u^2} \big|_{u=0} = 10395.$$

Also, $W(t) = \sqrt{t}W(1) \sim \mathcal{N}(0, t)$. Hence,

$$E[W^{12}(t)] = E[t^6 W^{12}(1)] = t^6 E[W^{12}(1)] = 10395t^6.$$

Therefore,

$$\text{Var}[I(T)] = 10395 \int_0^T t^6 dt = 1485T^7.$$

and

$$\text{Var}[I(\frac{1}{2})] = \frac{1485}{128} \approx 11.602.$$

The Monte Carlo simulation, as in Figure 1, gives us estimates $E(I(T)) = 6.162024\text{e-}03$ and $\text{Var}(I(T))=5.695168$, which are comparable to the theoretic values above.

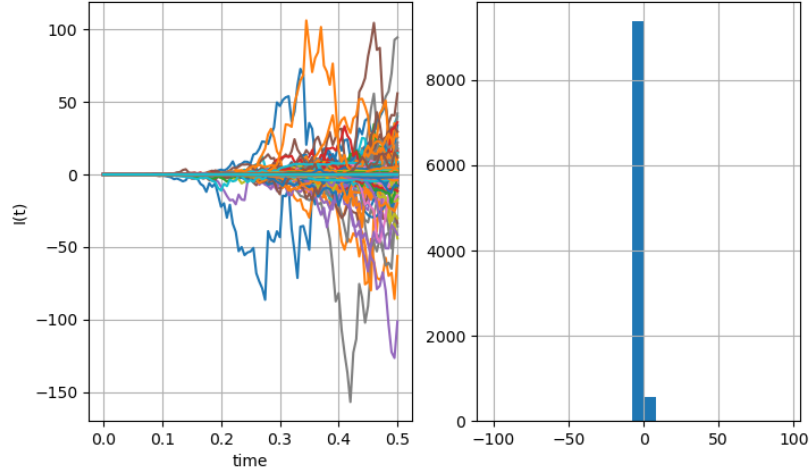


Figure 1: Results.

Exercise 2

a.

The moment-matching technique yields that the valuation of the European options under GBM with a time-dependent volatility can be performed by using a constant volatility coefficient

$$\sigma_* = \sqrt{\frac{1}{T} \int_0^T \sigma^2(t) dt}.$$

Hence, by (3.23), the analytical solution for the call option price is

$$V_c(t, S) = S(t)F_{\mathcal{N}(0,1)}(d_1) - Ke^{-r(T-t)}F_{\mathcal{N}(0,1)}(d_2),$$

where

$$\begin{aligned} d_1 &= \frac{\log\left(\frac{S(t)}{K}\right) + \left(r + \frac{1}{2}\sigma_*^2\right)(T-t)}{\sigma_*\sqrt{T-t}}, \\ d_2 &= \frac{\log\left(\frac{S(t)}{K}\right) + \left(r - \frac{1}{2}\sigma_*^2\right)(T-t)}{\sigma_*\sqrt{T-t}} = d_1 - \sigma_*\sqrt{T-t}, \\ S(t) &= S_0 \exp\left(\left(r - \frac{1}{2}\sigma_*^2\right)(t-t_0) + \sigma_*(W^Q(t) - W^Q(t_0))\right) \end{aligned}$$

b.

Using the model with the time-dependent volatility, we simulated the implied volatility (orange-line) in Figure 2. The blue line is σ^* .

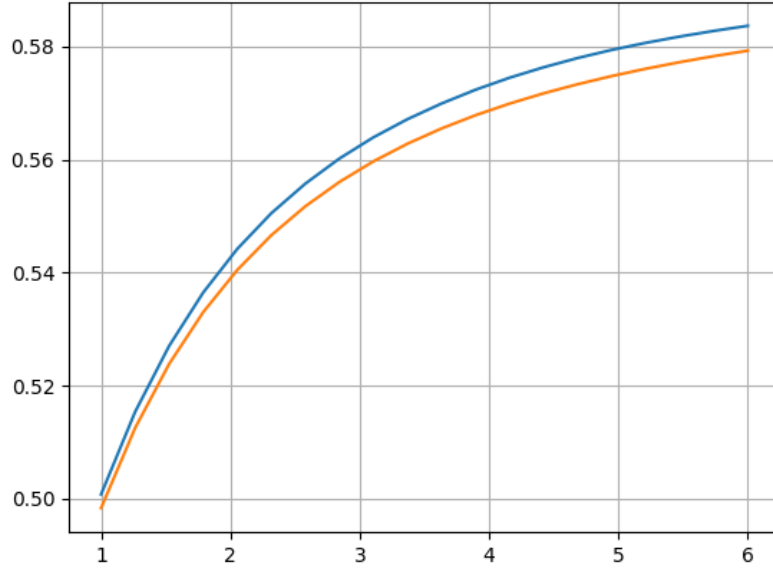


Figure 2: Results.

Exercise 3

a.

Since $S(t)$ satisfies (7.54), we have an AD process. Hence, by (7.55), the discounted characteristic function is of the following form

$$\phi_S(u; t, T) = e^{A(u, \tau) + B(u, \tau)S(t)}$$

where $\tau = T - t$, $A(u, 0) = 0$, and $B(u, 0) = iu$.

We obtain Riccati ODEs

$$\begin{aligned} \frac{dA}{d\tau} &= 0, \\ \frac{dB}{d\tau} &= \frac{\gamma^2}{2} B^2. \end{aligned}$$

Hence, the coefficients are $A = 0$ and

$$\int_0^\tau \frac{1}{B^2} dB = \frac{1}{2} \gamma^2 \tau,$$

so

$$B(u, \tau) = \frac{iu}{1 - \frac{1}{2} iu\tau\gamma^2}.$$

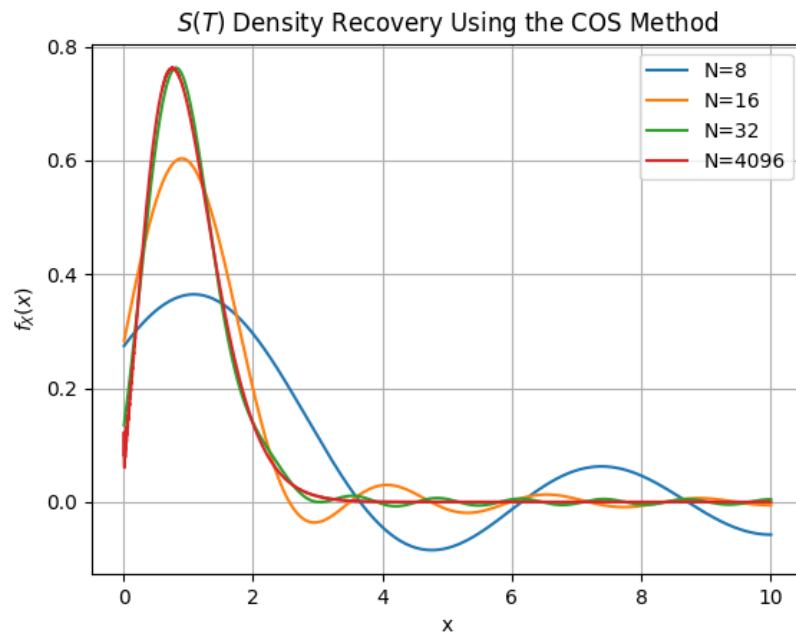
Therefore,

$$\phi_S(u; t, T) = \exp \left(\frac{iu}{1 - \frac{1}{2} iut\gamma^2} S(t) \right).$$

b.

Use (6.17) to approximate the density where $\bar{F}_k$ are given by (6.15). We simulate the PDF with Monte Carlo.

0.5



0.5

